

# GeoFusionNet / GeoCorr: Project Status Report

Status as of 2026-10-09, 19:20 UTC · main @ e92a340 · Prepared for Gobika S

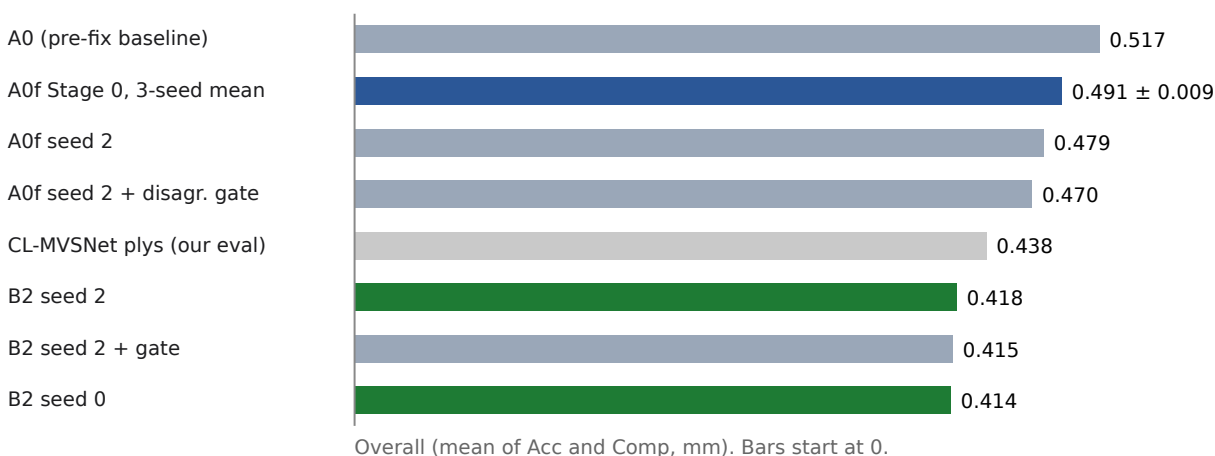
**Summary.** The frozen Stage 0 baseline (A0f) scores **0.491 ± 0.009** overall on full-res DTU (3 seeds × 22 scans). The geometric prior does not help in any of the ways tested: as an input (B1), as a loss (L\_geo), or as a confidence-gated replacement. In every case the effect is within noise or limited to a few outlier scans. Only one change has clearly helped so far: more cost-volume capacity (B2). It reaches **0.414 / 0.418** on two seeds, against A0f's 0.479 on seed 2. The gain comes from completeness, and per-view depth metrics did not improve. The third B2 seed, the B2 no-L\_geo ablation and a frozen mono-feature variant are running or queued now.

## 1. Ongoing work

| Queue / job                                                                                                      | State                                                                                                                                                                                                                                        | ETA                                           |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>B2 seeds + no-geo</b> (PID 3390614)<br>scripts/queue_b2_seeds_nogeo.sh ·<br>log runs/queue_b2_seeds_nogeo.log | B2_capacity seed0: <b>done</b> (overall 0.414)<br>B2_nogeo seed2: <b>running</b> , base phase step 18,900 of<br>20,000; then 6k fine-tune, 1.5k final phase, diagnostics,<br>probe, 1600×1152 eval<br>B2_capacity seed1: queued (full chain) | Queue<br>done<br>~2026-10-<br>11 07:30<br>UTC |
| <b>B2_mono_feat</b> (PID 3457546)<br>scripts/queue_b2_mono_feat.sh ·<br>patch fc8c805                            | Waiting for QUEUE_DONE above. Then seed 2: identity test,<br>train from B2_nogeo_finetune seed2, diagnostics, probe,<br>eval. Matched control: B2_nogeo seed2.                                                                               | Starts<br>~2026-10-<br>11 07:30<br>UTC        |
| B2 3-seed aggregate                                                                                              | scripts/aggregate_b2_3seed.sh (c54dfc2). Runs only after<br>QUEUE_DONE and writes<br>runs/B2_capacity/aggregated_1600x1152.json.                                                                                                             | After<br>queue                                |

Every queue step started after commit e92a340 (2026-10-09 ~17:10 UTC) logs a [provenance] git commit ... line.

## 2. Headline numbers (DTU test, 22 scans, 1600×1152, official protocol; mm, lower is better)



| Model                        | Acc                       | Comp          | Overall              | Notes                                                                |
|------------------------------|---------------------------|---------------|----------------------|----------------------------------------------------------------------|
| A0 (pre-fix)                 | 0.447                     | 0.586         | 0.517                | Superseded                                                           |
| <b>A0f Stage 0</b> (3 seeds) | 0.437 ± 0.003             | 0.545 ± 0.016 | <b>0.491 ± 0.009</b> | Frozen baseline (std over seeds, ddof=0)                             |
| A0f seed 2                   | 0.435                     | 0.523         | 0.479                | Reference for single-seed tests                                      |
| CL-MVSNet released plys      |                           |               | 0.438                | Scored with our evaluator (paper reports 0.329 under its own fusion) |
| B2_capacity seed 2           | 0.473                     | 0.362         | 0.418                | Better than A0f s2 on 17/22 scans                                    |
| B2_capacity seed 0           | 0.470                     | 0.358         | 0.414                | New, 2026-10-09                                                      |
| B2_capacity seed 1           | Not measured yet (queued) |               |                      |                                                                      |

### 3. How we got here

#### 3.1 Fixing the pipeline (2026-09-23 to 09-25)

- **Constant-depth failure:** early A0 runs had ~67 mm error. The cause was edge-aware smoothness applied to raw mm depth. Normalizing by per-image mean depth fixed it. A wrong score sign was ruled out.
- **Evaluation:** evaluation before 2026-09-23 had no fusion filtering and used a non-official metric. It now uses GeoMVSNet's open3d fusion (prob 0.3, 0.2 mm, 4 views) and the official DTUeval protocol, cross-checked against the reference. Full resolution is 1600×1152. **Numbers from before the fix are not comparable.**
- **Undertraining:** A0 saw ~40k samples, about 9% of the reference recipe. A CL-MVSNet-style LR fine-tune helped only ~5% (val fine median 1.75 → 1.65–1.70 mm), so the LR schedule does not explain the plateau.
- **Cost-volume fixes:** out-of-view masking plus trilinear regularizer upsampling cut sawtooth from 41% to 0.1%. They are now the default.

#### 3.2 Failure analysis: scan48 and the outlier scans (11, 13, 48, 62, 77)

- The main failure is **photometric ambiguity**. On wrong pixels, GT is the global cost minimum on only 30%, a secondary minimum on 41%, and no minimum on 28%. Our photometric loss itself prefers the wrong depth.
- Robust losses (L0.5, min-view) and best-k view aggregation did not fix it. Widening the depth windows at test time made it worse.
- The CL-MVSNet consistency loss (ICC + SCC, ported with ×0.0163 weight calibration) was within noise on aggregate val. On scan48 it was worse at the final checkpoint (4.34 vs 3.45 mm). **Dropped;** A0f (fixes only) was frozen as Stage 0.

### 3.3 Geometric prior G (Depth-Anything-V2-S anchored to SIFT points)

| Test                                        | Result (seed 2, matched control)                                                                                                                                      | Verdict                                  |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>B1:</b> G as input (zero-init GGF)       | Val fine +0.012 mm, coverage −1.2 pt; scan48 −0.44 mm, but the control's own swing between checkpoints is 0.39 mm; competing dips unchanged (18.53 vs 18.42%)         | No gain                                  |
| <b>No-L_geo</b> vs control                  | Val fine +0.025 mm; scan48 −0.20 mm; dips +0.15 pt                                                                                                                    | Noise, so L_geo has no measurable effect |
| Prior quality (22 scans)                    | Median  G−GT  5–69 mm vs model 0.8–4 mm. On ambiguous pixels G is nearer the true dip on only 52.4%, and within 4 mm on only 11.5%. 16/49 scan48 views have no prior. | About a coin flip as a dip selector      |
| Confidence gates (thresholds frozen on val) | Only "prior disagrees with coarse depth by $\geq 99.3$ mm" passes the pre-registered rule (test: 21.2% coverage, 86.7% correct). Partly mechanical.                   | Fixes gross outliers only                |
| Gated fusion, A0f s2 (22 scans)             | 0.479 → 0.470. Almost all of the gain is from scan48 (−0.116) and scan77 (−0.114); the other 20 scans are slightly worse.                                             | Small, concentrated                      |
| Gated fusion, B2 s2 (refit, 89.3 mm)        | 0.418 → 0.415. scan48 0.708 → 0.706.                                                                                                                                  | Near no-op on B2                         |

### 3.4 Capacity: B2\_capacity

- Changes: FPN 32/96/192, 8-group correlation cost, 3D UNet regularizer. 2.03M params vs A0f's 121k. Same 20k + 6k + 1.5k training chain as A0f.
- Fused point clouds are much better (comp 0.36 vs 0.52; scan48 1.213 → 0.708). **Per-view depth metrics are worse**, though: val fine median 1.76 vs 1.52 mm, and scan48 fine median 3.96 vs 3.45 mm. Competing dips on scan48 fell from 18.75 to 12.90%.
- Fusion pass-through check: the confidence filter is close to a no-op, and the point counts are about the same as A0f's (22.3M vs 22.1M). At a pass rate matched to A0f, B2 still scores 0.416. So the gain comes from *where* the points land, not how many survive fusion. The exact mechanism has not been verified.

## 4. Open questions and next decisions

- **Is B2's gain robust?** Two of three seeds agree (0.414, 0.418), and the seed 1 result is due ~10-11. The decision on the 3-seed mean  $\pm$  std comes from aggregate\_b2\_3seed.sh.
- **Does L\_geo matter on B2?** Answered by the B2\_nogeo seed 2 vs B2 seed 2 matched pair.
- **Can monocular information help without the noisy metric prior?** Answered by B2\_mono\_feat (frozen DA-V2-S features as a zero-init residual) vs B2\_nogeo seed 2.
- **GeoCorr Lite decision (still open):** the evidence so far does not support the prior as the paper's main contribution. Capacity and monocular features are the remaining candidates.
- Why B2 fuses better despite worse per-view depth is not yet explained.

Sources: runs/\*/eval\_1600x1152.json, runs/A0f\_fixes\_only\_short/aggregated\_1600x1152.json, matched-pair diagnostics and probes under runs/, NOTES.md and the experiment-history doc. All numbers are measured; items not yet run are marked "not measured".